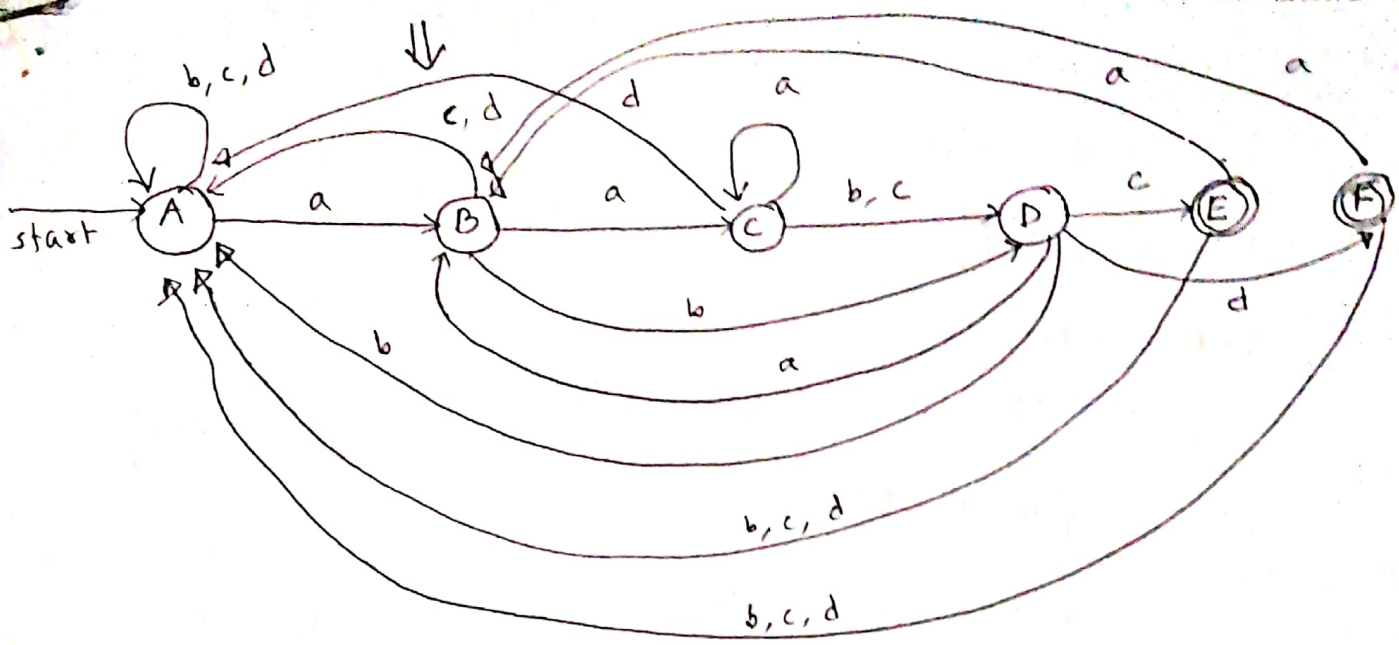
Lecture-2



Regular expressions:

Regular expressions denote language.

Example:

Variable declaration regular expression:

$$L(L|D)^*$$

Language: A variable should be started with L, then any number of L or D may be occurred.

Three operations on Languages:

1. Union
2. Concatenation
3. closure

Union:

Two languages L and M

Let $L = \{001, 10, 111\}$

$M = \{\epsilon, 001\}$

$$L \cup M = \{001, 10, 111\} \cup \{\epsilon, 001\}$$

$$= \{\epsilon, 10, 001, 111\}$$

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Concatenation:

$L = \{001, 10, 111\}$

$M = \{\epsilon, 001\}$

$L \cdot M = LM$

$= \{001, 10, 111\}\ \{\epsilon, 001\}$

$= \{001\epsilon, 10\epsilon, 111\epsilon, 001001, 10001, 111001\}$

$= \{001, 10, 111, 001001, 10001, 111001\}$

Closure:

$L = \{0, 1\}$

$L^* = \{\epsilon, 0, 1, 01, 10, \ldots\}$

regular expressions • inductive steps

Union: If E and F are regular expressions,

E+F is a regular expression denotes the union of L(E) and L(F) That is, L(E+F) = L(E)UL(F

Concatenation:

If E and F are regular expressions,

EF is a regular expression denotes the concatenate -nation of L(E) and E(F). That is L(EF) = L(E) L(F).

closure

If E is a regular expression, then E^* is a regular expression, denotes the closure of L(E). That is, $L(E^*) = (L(E))^*$

Parenthesised

If E is a regular expression, (E) is also a regular expression. Both E and (E) denote same language, that is L((E)) = L(E).

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